

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED****Q.1. Answer the following short questions.****(15x2=30)**

1. Define sample space.
2. If $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \cap B) = 0.15$, find $P(A|\bar{B})$
3. What do you mean by expected value of a random variable?
4. Under what conditions Poisson distribution is a limiting case of binomial distribution?
5. Given that X has a Poisson distribution with $P(X = 0) = 0.2$, find $P(X = 1)$
6. Write a short note on standard normal distribution.
7. What is the difference between a permutation and a combination?
8. How is an event of a random experiment defined?
9. What are equally likely events? Give example.
10. What is difference between independent and dependent events?
11. Why cannot mutually exclusive events also be independent?
12. For what value of A the function $f(x) = A(4x - 2x^2)$ for $0 \leq x \leq 2$ is a valid pdf?
13. If X is normally distributed with $\mu = 24$ and $\sigma = 4$, then find the 33rd percentile.
14. State multiplication law of probability for two independent events A and B .
15. In a normal distribution lower and upper quartiles are 28 and 55 respectively, find mean and standard deviation.

Q.2. Answer the following question.**(6x5=30)**

1. A continuous random variable X has the probability density function $f(x) = 20x^3(1-x)$ for $0 \leq x \leq 1$ and zero elsewhere. Find distribution function of X . Hence or otherwise find $P\left(\frac{1}{4} < X < \frac{1}{2}\right)$
2. The lifetime of a certain type of battery can be closely approximated by the normal curve with a mean of 350 hours and a standard deviation of 50 hours.
(i) What percentage of these batteries will have lifetime of more than 375 hours?
(ii) Above what value will the best ten percent of the batteries lie?
3. Ten vegetable cans, all the same size, have lost their labels. It is known that 5 contain tomatoes and 5 contain corn. If five are selected at random, what is the probability that all contain tomatoes? What is the probability that 3 or more contain tomatoes?
4. The experience of a house-agent indicates that he can provide suitable accommodation for 75 percent of the clients who come to him. If on a particular occasion 6 clients approach him independently, calculate the probability that
(i) exactly 4 clients, (ii) at least 5 clients, will get satisfactory accommodation.
5. Show that the mean and the variance of a Poisson distribution are equal.
6. The length of life for an automatic dishwasher is approximately normally distributed with a mean of 3.5 years and a standard deviation of 1.0 year. If this type of dishwasher is guaranteed for 12 months, what fraction of sales will require replacement?

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Q.1. Solve the following.

(6x5=30)

- (i) Analyze the conic represented by $16x^2 + 24xy + 9y^2 - 5x - 10y + 1 = 0$.
- (ii) Find the condition that the curves $ax^2 + by^2 = 1$, and $a_1x^2 + b_1y^2 = 1$ should intersect orthogonally.
- (iii) Find the asymptotes of the curve $(x - y + 1)(x - y - 2)(x + y) = 8x - 1$.
- (iv) Discuss the nature of the singular point $(0, 0)$ at the curve $(x^2 + y^2)^2 = 4a^2xy$.
- (v) Find equations of the straight line passing through the point $P(2, 0, -2)$ and perpendicular to each of the straight lines $\frac{x-3}{2} = \frac{y}{2} = \frac{z-1}{2}$ and $\frac{x}{3} = \frac{y+1}{-1} = \frac{z+2}{2}$.

Q.2. Solve the following.

(3x10=30)

- (a) Find pedal equation of the curve $r = a + b \cos \theta$.
- (b) Find the points at which the cardioid $r = 1 + \sin \theta$ has vertical tangents.
- (c) Find the area of the region bounded by one loop of the curve $r = a \sin 2\theta$. Also find the area of the region bounded by the circle $r = a$ and the curve $r = a \sin 2\theta$.
- (d) Show that the intrinsic equation of $3ay^2 = 2x^3$ is $9s = 4a(\sec^3 \alpha - 1)$.
- (e) Find equation of the sphere which passes through the circle $x^2 + y^2 + z^2 = 9$, $2x + 3y + 4z = 5$ and the point $(1, 2, 3)$.

PUACP



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Second Semester – Spring 2024

Paper: Computer –II

Course Code: COMP-103

Roll No. [REDACTED]

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(15x2=30)

1. Define Data Independence.
2. Write down components of database system.
3. What is compound key?
4. Who is "end user" in Database systems?
5. Define cardinality.
6. What are the benefits of normalisation?
7. Define transitive dependency.
8. What is referential integrity?
9. What is 1st NF in database?
10. Write down the properties of a relation.
11. Illustrate metadata?
12. Write the ingredients of data modeling.
13. What is entity? Which symbol is used to represent entity in ERD model?
14. What is conceptual database design?
15. What we do in the analysis phase of database design process?

Answer the following questions:

(3x10=30)

- Q2: Draw and explain ERD of "order Processing System".
- Q3: What are the advantages of DBMS over traditional file system? Explain.
- Q4: Convert the following table to 3NF step by step.

DepartmentID	Dname	EmployeeID	Ename	ProductID	Pname	Btime
10	Finance	1, 5, 11	Hassan	27, 25, 22	Alpha, Beta, Gamma	4, 3, 7
14	R&D	2, 4	Jack, Samuel	26, 21	Pail, Hill	8, 9



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(6x5=30)

Q.1. Answer the following short questions

1. What is the role of Time Management in student's success?
2. Describe the Uses of Library.
3. How to remove hurdles in improving reading skills.
4. Explore your Anxieties as a student during the exams.
5. Differentiate between Hearing and Listening during the lecture.
6. What is the difference between Topic Sentence and Thesis Statement?

(3x10=30)

Q.2. Answer the following questions

1. Write a paragraph on anyone of the following topics: (Word Limit: 120)
 - a) Importance of Education
 - b) Role of Social Media
2. Write a brief note on Skimming and Scanning as techniques of developing reading skills.
3. Read the following passage carefully and answer the following questions.

Stress is a body reaction to any demands or changes in its internal and external environment. Whenever there is a change in the external environment such as temperature, pollutants, humidity and working conditions, it leads to stress. In these days of competition when a person makes up his mind to surpass what has been achieved by others, leading to an imbalance between demands and resources, it causes psycho-social stress. It is a part and parcel of everyday life. Stress has a different meaning, depending on the stage of life you are in. The loss of a toy or a reprimand from the parents might create a stress shock in a child. An adolescent who fails in the examination may feel as if everything has been lost and life has no further meaning. In an adult the loss of his or her companion, job or professional failure may appear as if there is nothing more to be achieved. Such signs appear in the attitude and behavior of the individual, as muscle tension in various parts of the body, palpitation and high blood pressure, indigestion and hyperacidity. Ultimately the result is self-destructive behavior such as eating and drinking too much, smoking excessively, relying on tranquilizers. There are other signs of stress such as trembling, shaking, nervous blinking, dryness of throat and mouth and difficulty in swallowing. In a study sponsored by World Health Organization and carried out by Harvard School of Public Health, the global burden of diseases and injury indicated that stress diseases and accidents are going to be the major killers in 2020. The heart disease and depression – both stress diseases – are going to rank first and second in 2020. Road traffic accidents are going to be the third largest killers. These accidents are also an indicator of psychosocial stress in a fast-moving society. Other stress diseases like ulcers, hypertension and sleeplessness have assumed epidemic proportions in modern societies. A person under stress reacts in different ways and the common ones are flight, fight and flee depending upon the nature of the stress and capabilities of the person. The three responses can be elegantly chosen to cope with the stress so that stress does not damage the system and become distress.

1. What is stress?
2. What factors lead to stress?
3. What are the signs by which a person can know that he is under stress?
4. What are the different diseases a person gets due to stress?
5. How does a person react under stress?



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Second Semester – Spring 2024

Paper: Pakistan Studies

Course Code: PST-111

Roll No. 12

Time: 3 Hrs.

30

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اس پرچہ کو مہیا کی گئی جوابی کاپی پر حل کریں۔

Q.1. Write short note on the following.

(6x5=30)

سوال نمبر 1۔ مندرجہ ذیل پر مختصر نوٹ تحریر کریں۔

i. Indus Basin Treaty

معادہ سندھ طاس

.i

ii. Tahreek-i-Mujahideen

تحریک مجاہدین

.ii

iii. Establishment of All India Muslim League

آل انڈیا مسلم لیگ کا قیام

.iii

iv. Cabinet Mission Plan

کابینہ مشن پلان

.iv

v. Shimla Agreement

شملا معاہدہ

.v

vi. Durand Line

ڈیورنڈ لائن

.vi

Q.2. Answer the following questions

(2x15=30)

سوال نمبر 2۔ مندرجہ ذیل سوالات کے جوابات تحریر کریں۔

i. Write a note on the causes of the Ayub Khan's Martial law.

1۔ ایوب خان مارشل لاء کے اسباب پر نوٹ لکھیں۔

ii. What do you know about the TWO-NATION theory?

2۔ آپ دو قومی نظریہ کے بارے میں کیا جانتے ہیں؟

7 یومی	8 الربع	9 مہین
10 آباؤکم	11 بنت الاخی	12 شقاق
13 بخل	14 السبیل راضی	15 مبین
16 تَاكُونُ	17 نَحْنُ	18 الَّذِينَ
19 عَامٌ	20 النَّاسُ	

(20x0.5=10)

Q.4. Write the plural with the given words.

سوال نمبر 4: مندرجہ ذیل الفاظ کی مدد سے واحد کی جمع لکھیں۔

فکلو - اَمْوَاتٌ - کُھم - صالحون - اَیْدِیْ - رُسُلٌ - کُتِبَ - عَافِلُونَ - شَیَاطِیْنٌ - یُؤْمِنُونَ - یُفْسِدُونَ - شَہداء - ابواب - حَسَنَات - اَلْبَاب - اَعْمَالٌ - یَتْلُو - مَؤْمِنُونَ - اَسْمَاءٌ - اَصْحَاب

یؤمن	هُوَ	یُفْسِدُ	کُتِبَ	بَابٌ
فَعَلَ	شہید	صالح	عَافِلٌ	حَسَنَةٌ
مَوْتُ	مُؤْمِنٌ	یَدٌ	صَاحِبٌ	لُبٌ
یَتَنِمُّ	عَمَلٌ	رَسُولٌ	شیطان	شَیَاطِیْنٌ

Q.5. Match the verses with lines to the translation.

(5x2=10)

سوال نمبر 5: آیات کو ان کے ترجمے سے لکیر کے ذریعے ملائیں۔

1	وَلَا تَلْسُوا الْحَقَّ بِالْبَاطِلِ وَتَكْتَفُوا الْحَقَّ وَأَنْتُمْ تَعْلَمُونَ	اس کی طرف سے (ان کے لئے بہت) درجات ہیں اور ہمیشہ اور رحمت ہے، اور اللہ بڑا بخشنے والا مہربان ہے
2	فَرَجَبٌ مِّنْهُ وَ مَغْفِرَةٌ وَ رَحْمَةٌ وَ كَانَ اللَّهُ غَفُورًا رَّحِيمًا	اس نے کہا میں اللہ کی پناہ میں آتا ہوں کہ میں جاہلوں میں سے ہو جاؤں۔
3	قَالَ أَعُوذُ بِاللَّهِ أَنْ أَكُونَ مِنَ الْجَاهِلِينَ	(یہ) وہ کتاب ہے جس میں کوئی شک نہیں، پر ہیزگاروں کے لیے ہدایت ہے۔
4	وَ إِذَا صَلَّيْتُمْ فَبِ الْأَرْضِ فَلْيَسْنَ عَلَيْكُمْ جُنَاحٌ أَنْ تَقْصُرُوا مِنَ الصَّلَاةِ	اور حق کو باطل کے ساتھ غلط ملکہ کر دو، اور حق کو (ان) چھپاؤ حالانکہ تم جانتے ہو۔
5	ذَلِكَ الْكِتَابُ لَا رَيْبَ فِيهِ هُدًى لِّلْمُتَّقِينَ	اور جب تم زمین میں سفر کرو تو تم پر کوئی سنا نہیں کہ تم نماز میں قصر کرو۔